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Second Order Correlation

Speaker: Zhongqi Xiu

OUTLINE

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1. Second correlation function

2. Examples of different light fields

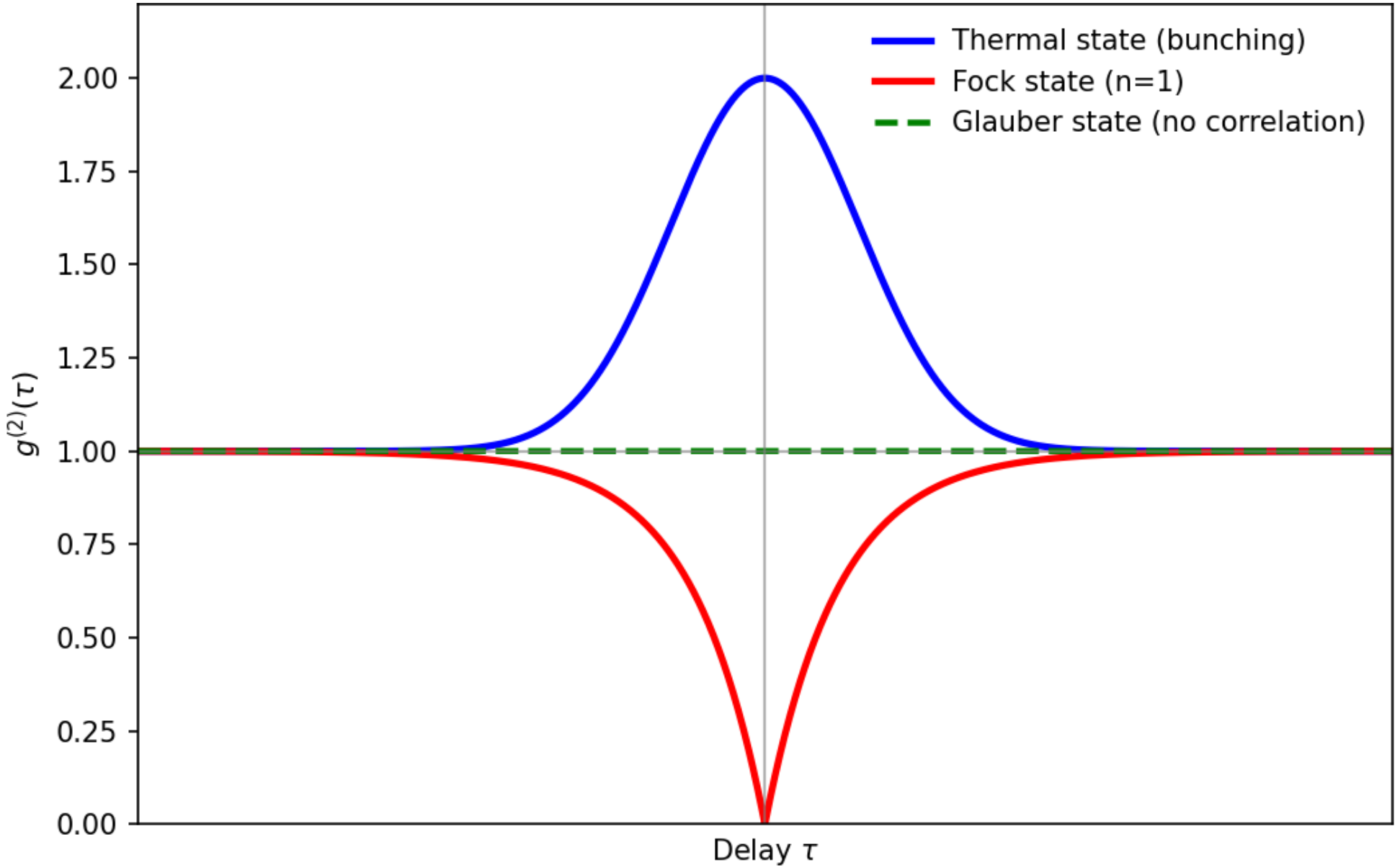
3. Measurement and data analyze

01.

Correlation Functions

Single photon sources

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Classical correlation function

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Correlation between two separated intensity signals

$$g^{(2)}_{class.}(\tau) = \frac{\langle I(t)\, I(t+\tau)\rangle}{\langle I(t)\rangle^2} = \frac{\langle E^*(t)\, E^*(t+\tau)\, E(t+\tau)\, E(t)\rangle}{\langle E^*(t)\, E(t)\rangle^2}$$

Using the second quantization to transform field into quantum operators

$$\hat{E}_k(t) = \hat{E}_k^{(+)}(t) + \hat{E}_k^{(-)}(t)$$

$$\hat{E}_k^{(+)}(t) \propto \hat{a}_k \cdot \exp\left(-i\left(\omega_k t - \vec{k}\cdot\vec{r}\right)\right)$$

$$\hat{E}_k^{(-)}(t) \propto (\hat{a}_k)^\dagger \cdot \exp\left(+i\left(\omega_k t - \vec{k}\cdot\vec{r}\right)\right)$$

Quantum correlation function

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$$g_{QM}^{(2)}(\tau) = \frac{\left\langle \hat{E}_k^{(-)}(t)\, \hat{E}_k^{(-)}(t+\tau)\, \hat{E}_k^{(+)}(t+\tau)\, \hat{E}_k^{(+)}(t) \right\rangle}{\left\langle \hat{E}_k^{(-)}(t)\, \hat{E}_k^{(+)}(t) \right\rangle^2}$$

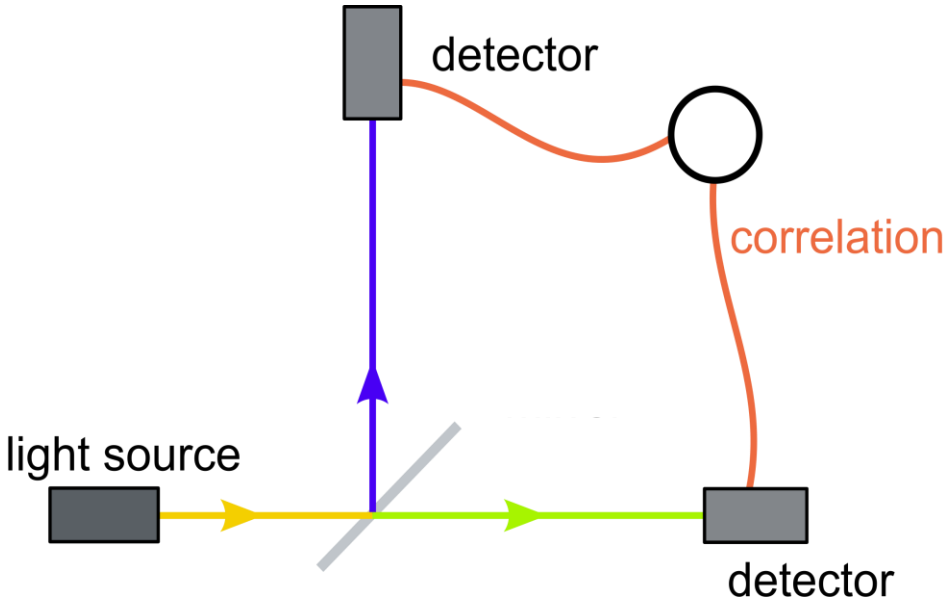
$$\overset{(\tau \to 0)}{=} \frac{\left\langle (\hat{a}_k)^\dagger (\hat{a}_k)^\dagger \hat{a}_k \hat{a}_k \right\rangle}{\left\langle (\hat{a}_k)^\dagger \hat{a}_k \right\rangle^2} = \frac{\langle n(n-1) \rangle}{\langle n \rangle^2}$$

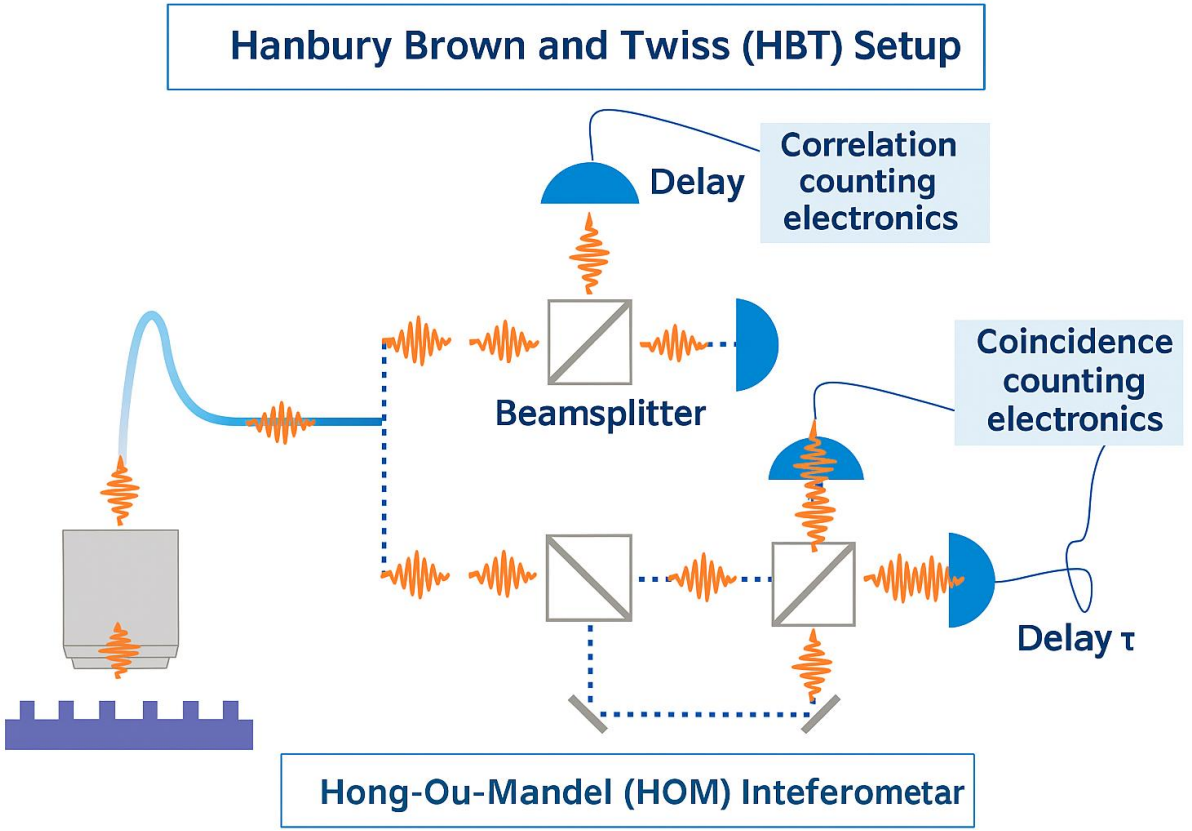
$$g_{QM}^{(2)}(0) = 1 + \frac{(\Delta n)^2 - \langle n \rangle}{\langle n \rangle^2}$$

Conditional probability: detect a second photon when one photon was already detected.

Typical HBT setup

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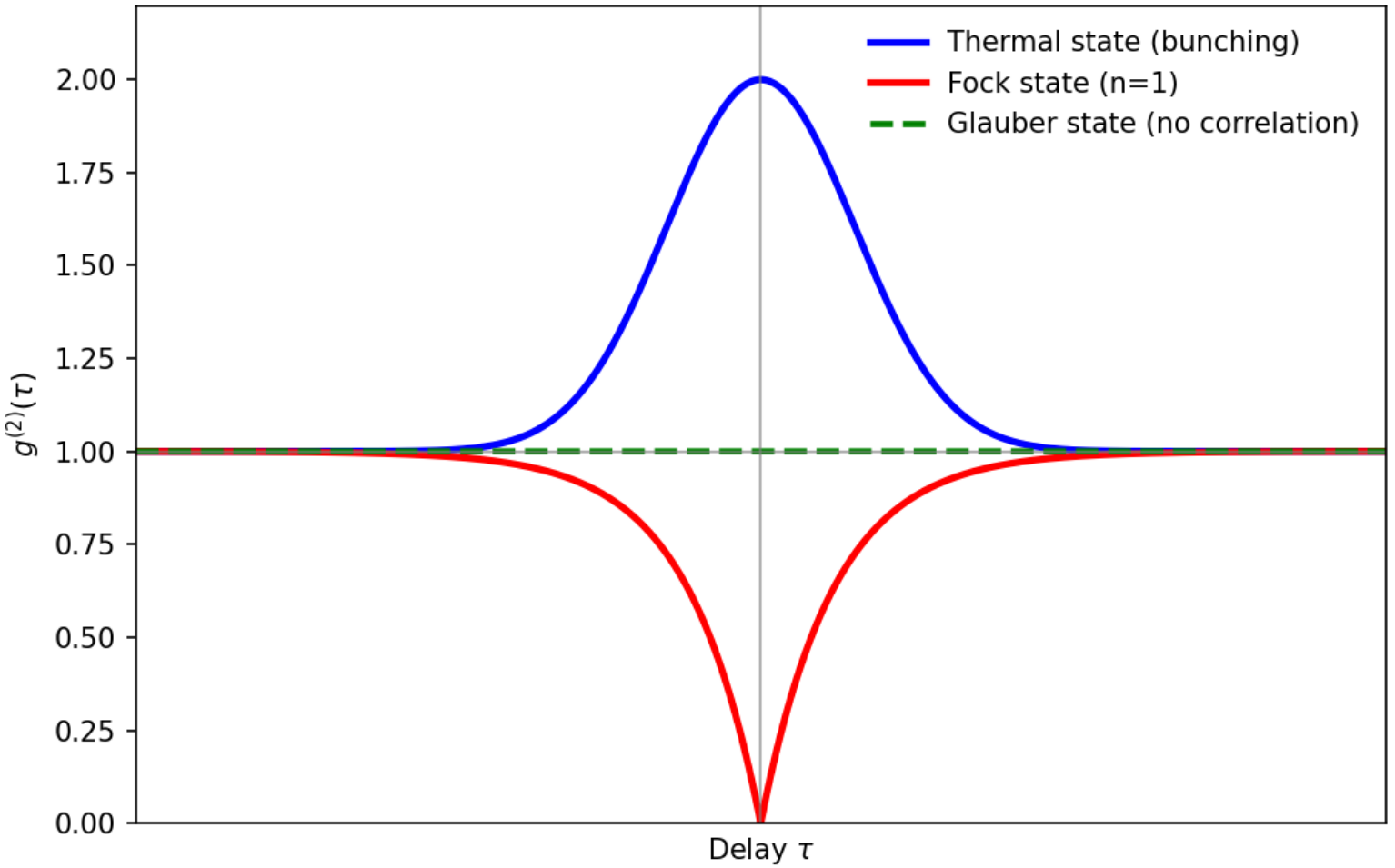
J. Phys. Chem. Lett. 2023, 14, 13, 3274–3284

02.

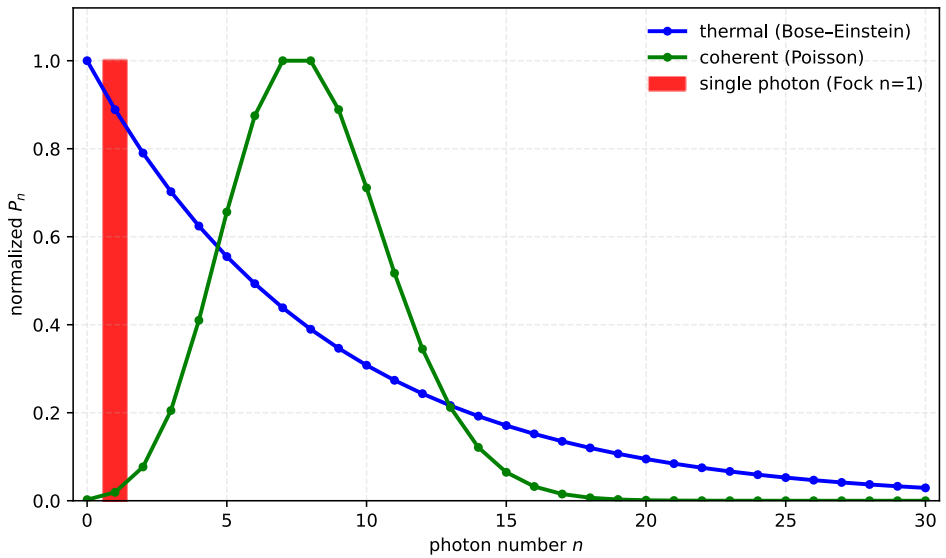
Examples of different light fields

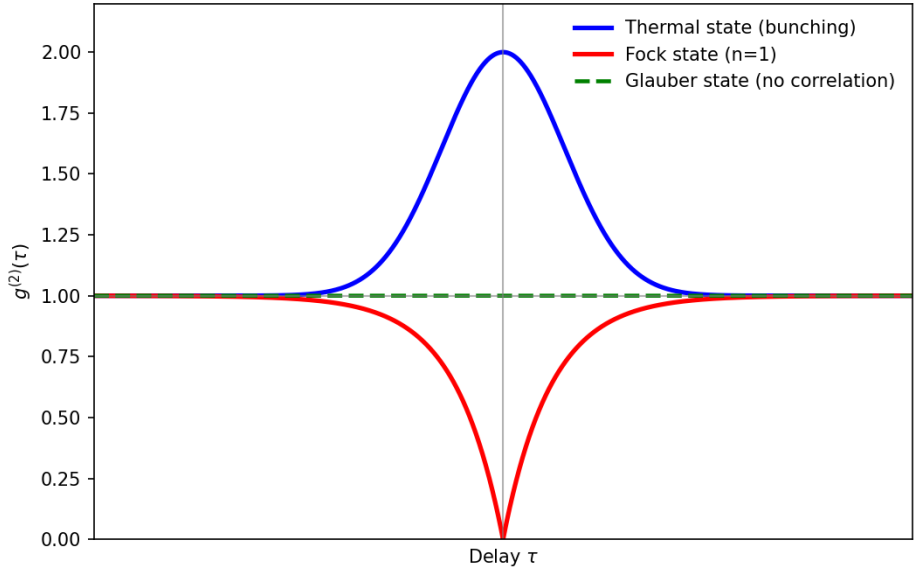
Ideal light fields

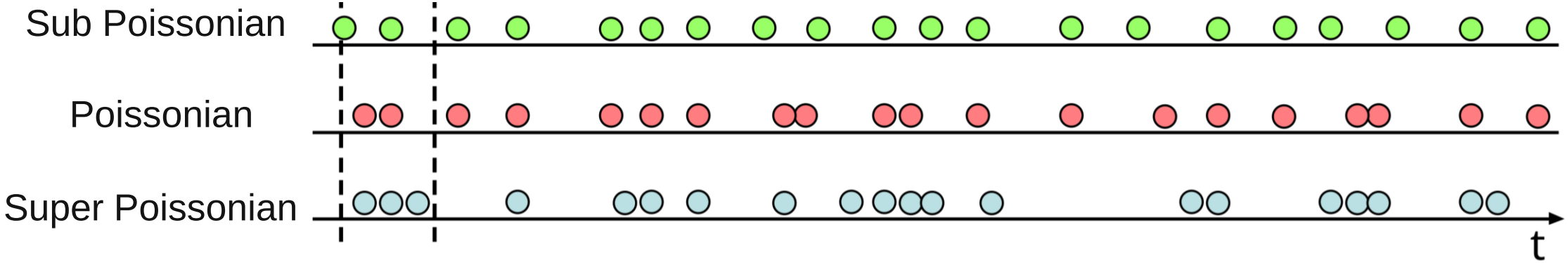
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Photon distribution





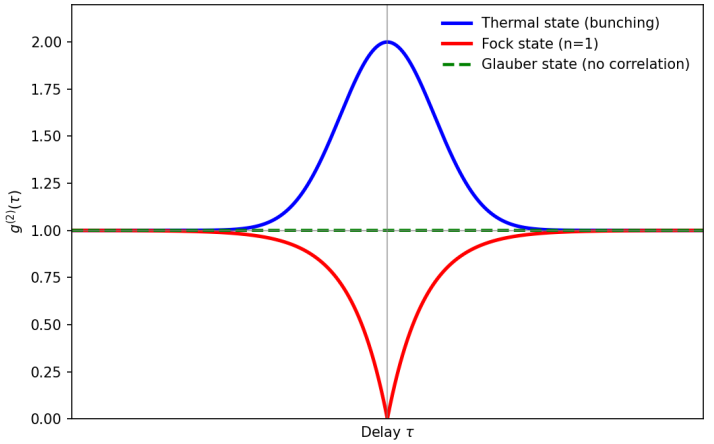


Thermal light

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Siegert relation $g^{(2)}(\tau) = 1 + \left|g^{(1)}(\tau)\right|^2$

$$g^{(1)}(\tau) = \frac{\langle E^*(t)E(t+\tau)\rangle}{\langle I(t)\rangle}$$



Time correlation relation in Gaussian and Lorentz line

$$|g^{(1)}(\tau)| = e^{-\pi\Delta\nu|\tau|} \quad \tau_c^{(\mathrm{L})} = 1/(\pi\Delta\nu)$$

$$|g^{(1)}(\tau)| = \exp\left(-2\pi^2\sigma^2\tau^2\right) \quad \tau_c^{(\mathrm{G})} = 1/(\sqrt{2}\pi\sigma) = \sqrt{2\ln 2}/(\pi\Delta\nu_{\mathrm{G}})$$

$$\Delta\nu \approx 2.7 \times 10^{11}\mathrm{Hz}, \tau_c \sim 1/(\pi\Delta\nu) \approx 1.2\mathrm{ps}$$

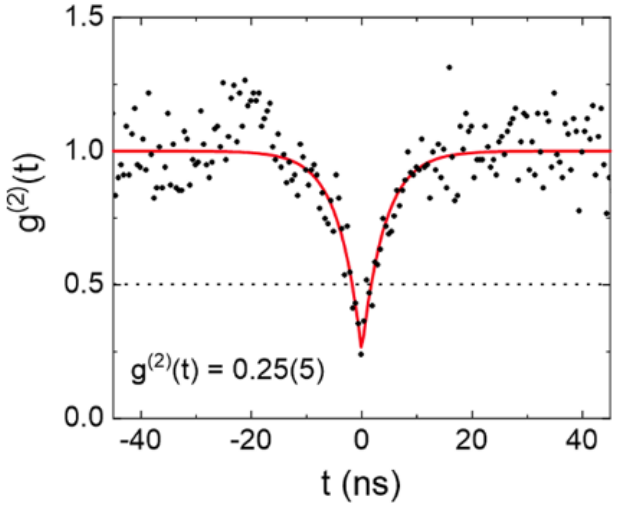
Laser (Coherent Light)

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Continue wave laser

For ideal Poisson distribution:

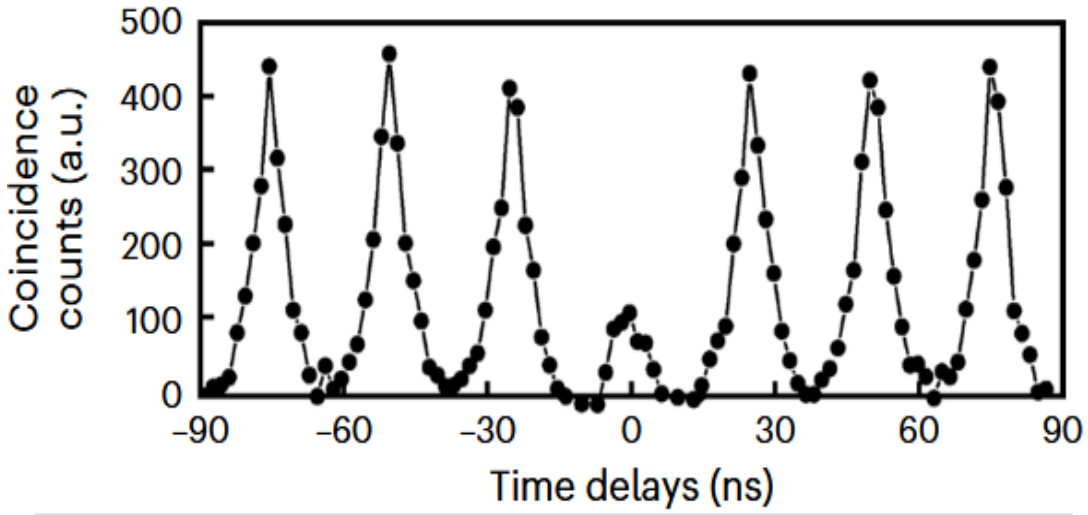
$$g^{(2)}(\tau) = \frac{\langle \hat{a}^\dagger(t)\hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\hat{a}(t)\rangle}{\langle \hat{a}^\dagger\hat{a}\rangle^2} = 1 \quad (\forall\tau)$$



Pulsed laser

$$I(t) = \sum_k s(t - kT_r)$$

$$G^2(\tau) \propto \int I(t)I(t+\tau)dt = \sum_{k\in\mathbb{Z}} \underbrace{\int s(t)s(t+\tau-kT_r)dt}_{\equiv R_0(\tau - kT_r)}$$



Nano Letter 2021, 21, 4715−4720
Nature Materials 2023, 22, 1311–1316

Fitting Function

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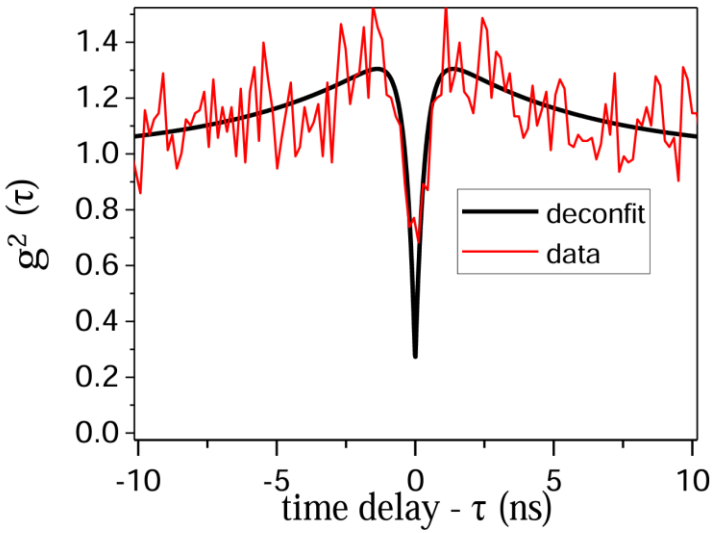
Continue wave laser

For typical two-level system:

$$g^{(2)}_{2-\text{level}}(\tau) = 1 - \exp[-(W+\Gamma)|\tau|] \qquad \Gamma = 1/T_1$$

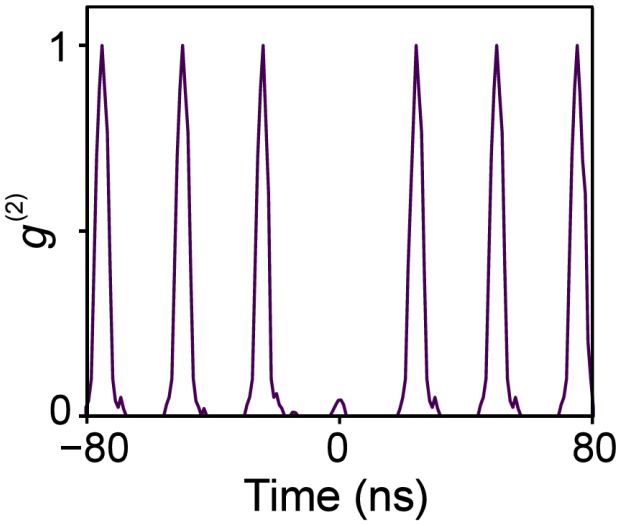
For solid emittor system: (detail in PRL paper)

$$g^{(2)}_{3-\text{level}}(\tau) = 1 - (1+a)e^{-|\tau|/\tau_1} + ae^{-|\tau|/\tau_2}$$



Pulsed laser

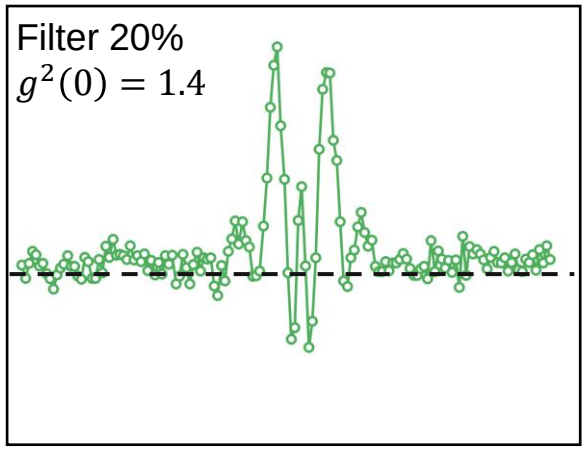
$$g^{(2)}(0) \approx \frac{\text{Area}(k=0)}{\frac{1}{2}\left[\text{Area}(k=+1) + \text{Area}(k=-1)\right]}$$

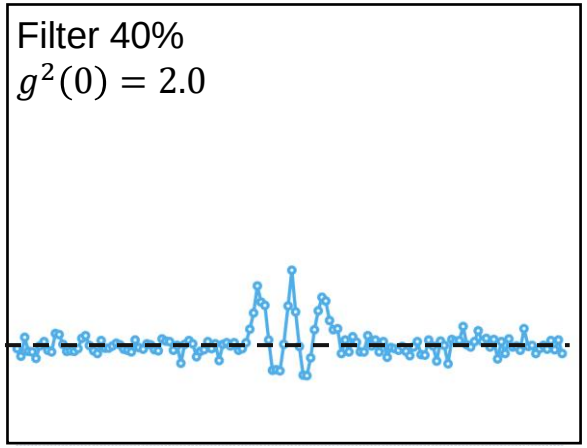


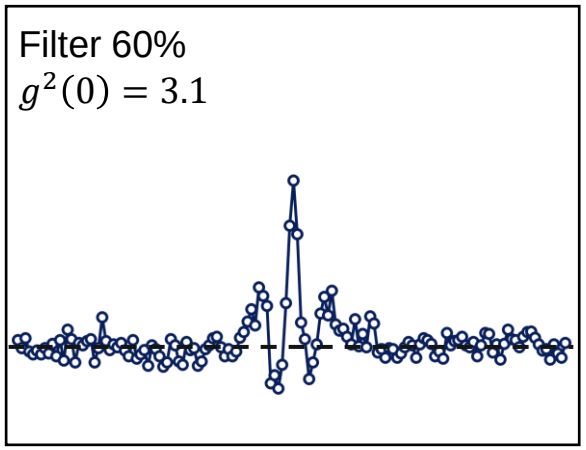
Phys. Rev. Lett. 85, 290, 2000

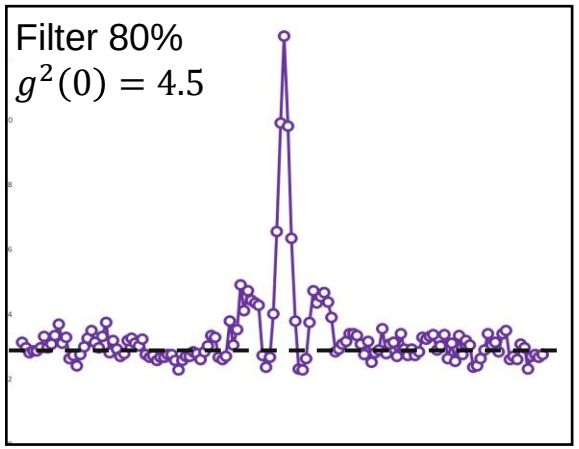
*Entangled photon pairs

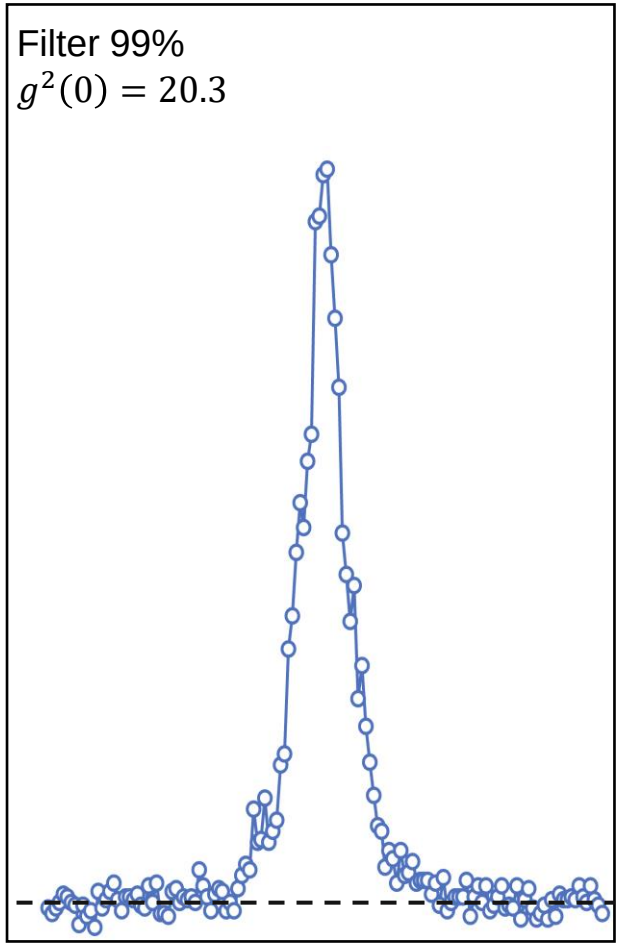
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$$\frac{\langle I(t)\, I(t+\tau)\rangle}{\langle I(t)\rangle^2}$$

Phys. Rev. Lett. 134, 053401

Multimode

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For white light:

$$g^{(2)}(0) = \frac{\langle n(n-1)\rangle}{\langle n\rangle^2} = 1 + \frac{(\Delta n)^2 - \langle n\rangle}{\langle n\rangle^2} \qquad g^{(1)}(\tau) = \frac{\langle E^*(t)E(t+\tau)\rangle}{\langle I(t)\rangle}$$

Negative binomial distribution for M modes: $\mathrm{Var}(n) = \bar{n} + \frac{\bar{n}^2}{M}$

$$g^{(2)}(0) = 1 + \sum_k w_k^2 = 1 + \frac{1}{M} \qquad w_k = \langle n_k\rangle / \langle n\rangle$$

For single photon emittors:

$$g^{(2)}(0) = \frac{\langle n(n-1)\rangle}{\langle n\rangle^2} = 1 + \frac{(\Delta n)^2 - \langle n\rangle}{\langle n\rangle^2} = 1 - \frac{1}{N}$$



Background correction

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Assume Independent and Uncorrelated Signal and Background

$$\begin{cases} I_1(t) = S_1(t) + B_1(t) \quad I_2(t) = S_2(t) + B_2(t) \\ \langle I_1 I_2 \rangle = \langle S_1 S_2 \rangle + \langle S_1 \rangle \langle B_2 \rangle + \langle B_1 \rangle \langle S_2 \rangle + \langle B_1 B_2 \rangle \\ \langle S_1 S_2 \rangle = \langle S_1 \rangle \langle S_2 \rangle g^{(2)}_{SS}(\tau) \quad \langle B_1 B_2 \rangle = \langle B_1 \rangle \langle B_2 \rangle g^{(2)}_{BB}(\tau) \end{cases} \qquad \rho_1 = \frac{\langle S_1 \rangle}{\langle I_1 \rangle} \quad \rho_2 = \frac{\langle S_2 \rangle}{\langle I_2 \rangle}$$

$$g^{(2)}_{\text{meas}}(\tau) = 1 + \rho_1 \rho_2 \left(g^{(2)}_{SS}(\tau) - 1 \right) + (1 - \rho_1)(1 - \rho_2) \left(g^{(2)}_{BB}(\tau) - 1 \right)$$

$$\Downarrow$$

$$g^{(2)}_{SS}(0) = \frac{g^{(2)}_{\text{meas}}(0) - (1 - \rho^2)}{\rho^2}$$

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03.

Measurement and data analyze

Detector: Excelitas SPCM-AQRH-WX-TR

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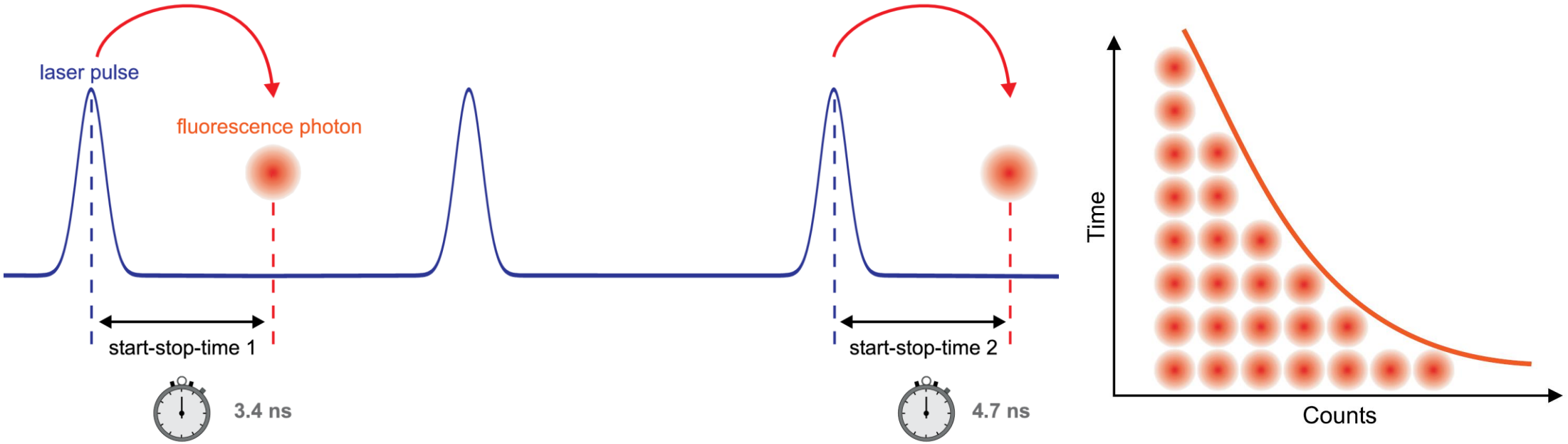
Order Part#	W - Output Pulse Options			
WX-YY	Output Pulse Width (ns)	Dead Time (ns)	Output Pulse Height (V)	-W0
SPCM-AQRH-1X-TR	10	22	2.2	≤1500 cps
SPCM-AQRH-2X-TR	18	28	2.2	≤1500 cps
SPCM-AQRH-3X-TR	28	35	2.2	≤1500 cps
SPCM-AQRH-4X-TR	10	22	4.4	≤1500 cps
SPCM-AQRH-5X-TR	18	28	4.4	≤1500 cps
SPCM-AQRH-6X-TR	28	35	4.4	≤1500 cps

time1.csv

channel	absTime
0	0.000001889075
0	0.000009640500
0	0.000012378775
1	0.000033345925
1	0.000065020045
0	0.000087585198
0	0.000101618498
1	0.000144418722
0	0.000186015226
1	0.000218612768
1	0.000246086740
0	0.000257210602
0	0.000291129370
1	0.000293879944
0	0.000298802741
1	0.000319446001
0	0.000327347137
1	0.000337240606

TCSPC (Time-Correlated Single Photon Counting)

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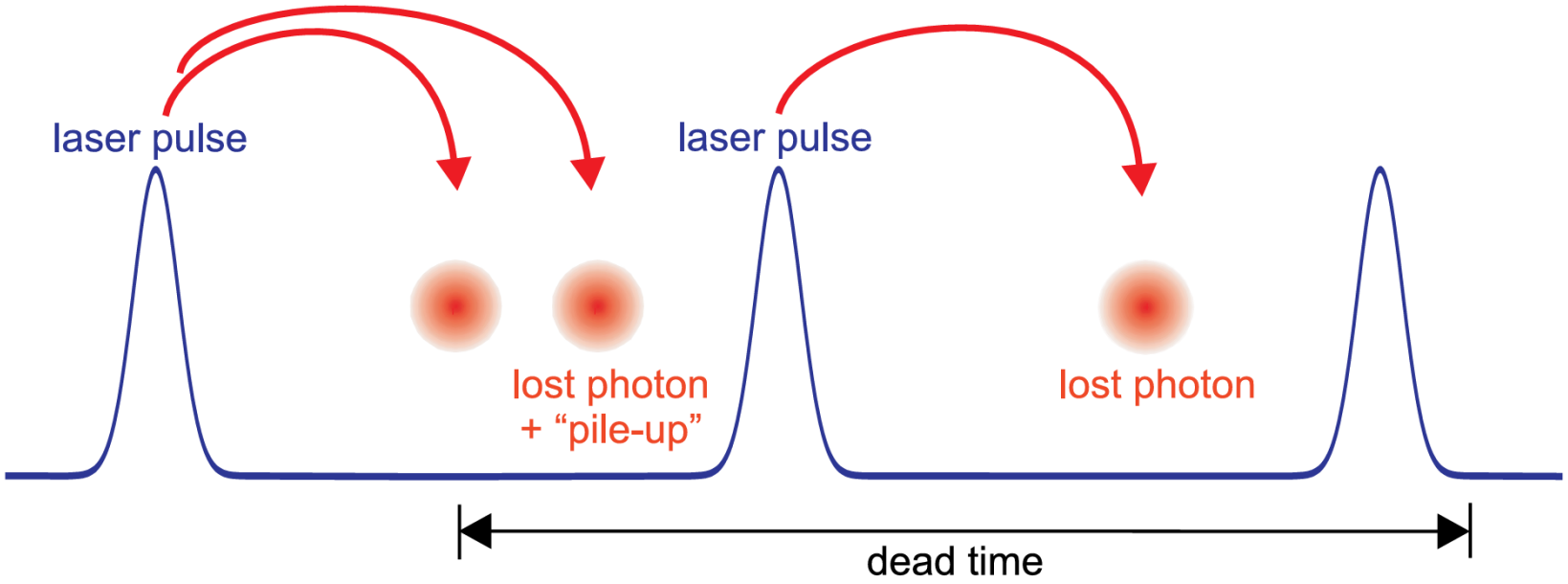


The stopwatch readings are sorted into a histogram consisting of a range of “time bins”.

The actual time difference measurement is done by means of fast electronics which provide a digital timing result.

TCSPC (Time-Correlated Single Photon Counting)

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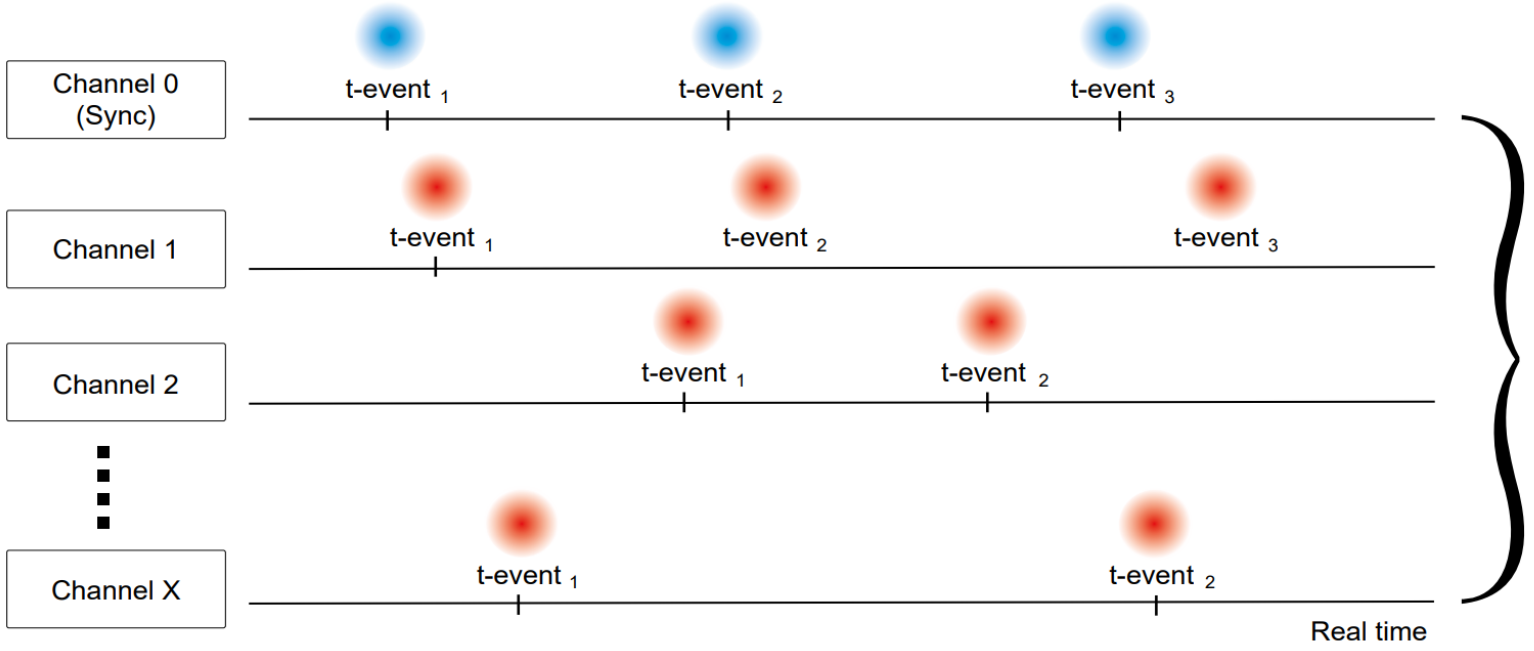


Detector and electronics have a “dead” time for at least some ns after a photon event.

TCSPC systems are designed to register only one photon per excitation cycle.

Measurement in T2 mode

Every input is treated as a photon channel



time1.csv

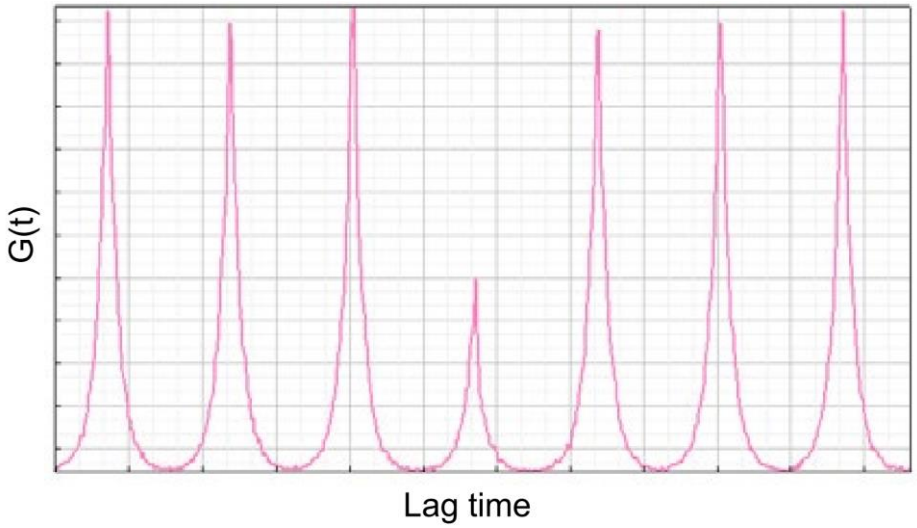
channel	absTime
0	0.000001889075
0	0.000009640500
0	0.000012378775
1	0.000033345925
1	0.000065020045
0	0.000087585198
0	0.000101618498
1	0.000144418722
0	0.000186015226
1	0.000218612768
1	0.000246086740
0	0.000257210602
0	0.000291129370
1	0.000293879944
0	0.000298802741
1	0.000319446001
0	0.000327347137
1	0.000337240606

The device simply time-stamps every event with its absolute arrival time (since acquisition start) at the hardware resolution (25 ps).

Correlator: Timeharp 260

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	TimeHarp 260
Detection channels*	1 or 2
Temporal resolution	25 ps (PICO) 250 ps (NANO)
Dead time	< 25 ns (PICO) < 2 ns (NANO)
Jitter**	< 20 ps (PICO) < 250 ps (NANO)



Normalize:

$$g^{(2)}(\tau) \approx \frac{G(\tau)}{R_A R_B T \Delta t}$$

$G(\tau)$ is the difference histogram in bin Δt

R_A, R_B are mean count rates over measurement time T

Data process

Timeharp 260 software TTTR mode

$$G_{XY}(\tau) = \frac{\langle \delta I_X(t)\ \delta I_Y(t+\tau)\rangle}{\langle I_X\rangle\langle I_Y\rangle}, \quad \delta I = I - \langle I\rangle$$

Self-correlation $$g^{(2)}_{AA}(\tau) = \frac{\langle I_A(t) I_A(t+\tau)\rangle}{\langle I_A\rangle^2} = 1 + G(A,A)$$

Cross-correlation $$g^{(2)}_{AB}(\tau) = \frac{\langle I_A(t)\ I_B(t+\tau)\rangle}{\langle I_A\rangle\ \langle I_B\rangle} = 1 + G(A,B)$$

Dead time & After pulsing effect:

$$G^{(\mathrm{Dead})}_{AA}(\tau) \approx \begin{cases} -1 & 0 \le |\tau| < \tau_d \\ 0 & |\tau| \ge \tau_d \end{cases}$$

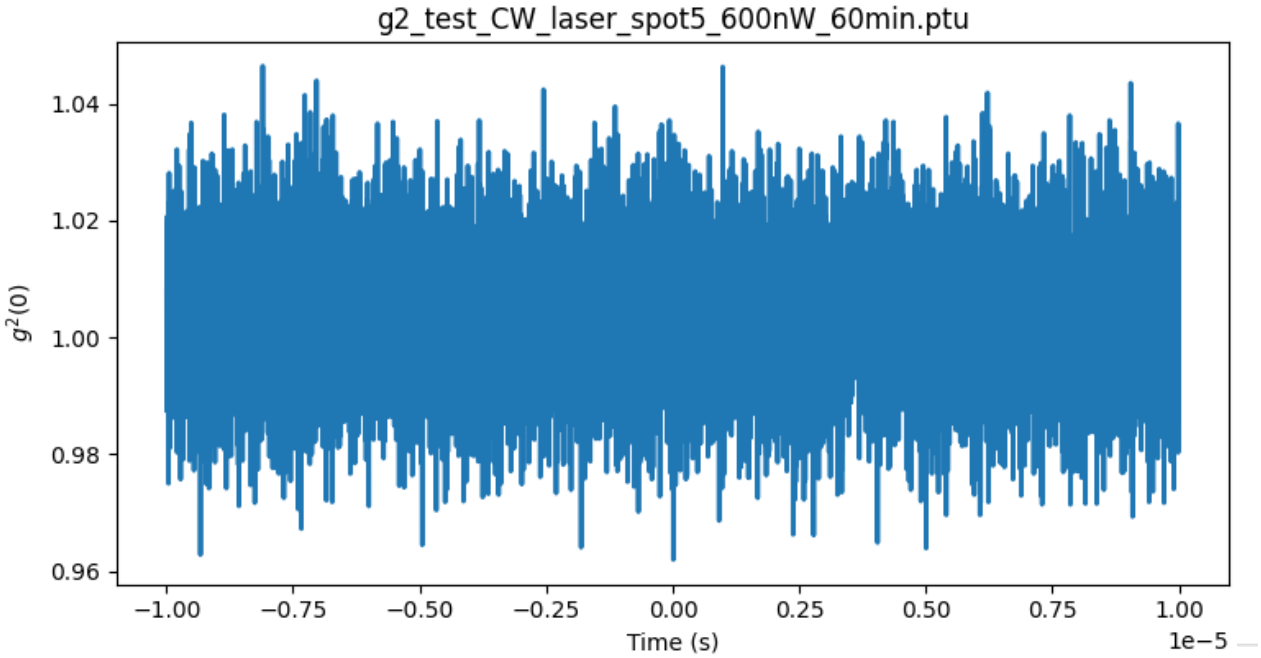
$$G^{(\mathrm{AP})}_{AA}(\tau) \approx \sum_i A_i e^{-|\tau|/\tau_{AP,i}} \quad A_i > 0$$

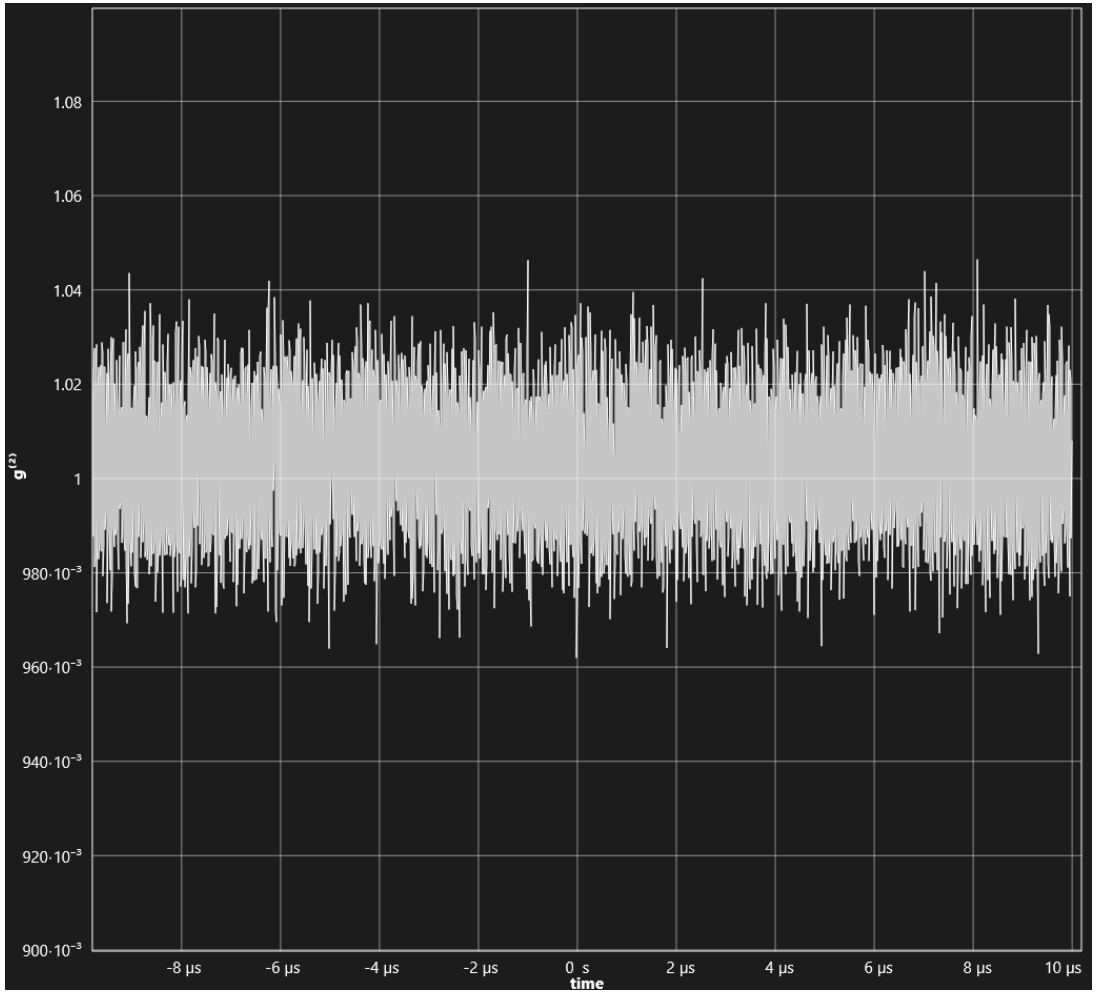
g2_test_pulsed_laser_spot2_1000nW.cor

taustep	tau/s	G(A,A)	G(B,B)	G(A,B)
5	0.0000001249	0.1606	0.1874	0.0380
6	0.0000001499	0.1308	0.1444	0.0454
7	0.0000001748	0.0874	0.1103	0.0345
9	0.0000002248	0.0926	0.0794	0.0397
11	0.0000002747	0.0685	0.0687	0.0295
13	0.0000003247	0.0447	0.0659	0.0326
15	0.0000003746	0.0395	0.0456	0.0341
17	0.0000004246	0.0347	0.0423	0.0249
19	0.0000004745	0.0389	0.0348	0.0154
21	0.0000005245	0.0316	0.0323	0.0182
1441783	0.0360085304	0.0000	0.0000	0.0001
1572855	0.0392820536	0.0000	0.0000	-0.0000
1703927	0.0425555768	-0.0000	0.0000	-0.0000
1834999	0.0458291000	-0.0000	-0.0000	0.0000
1966071	0.0491026232	0.0000	-0.0000	0.0000
2097143	0.0523761464	0.0000	0.0000	0.0000
2359287	0.0589231928	0.0000	0.0000	-0.0000
2621431	0.0654702392	0.0000	-0.0000	0.0000
2883575	0.0720172856	0.0000	0.0000	0.0000
3145719	0.0785643320	-0.0000	-0.0000	0.0000
3407863	0.0851113784	0.0000	0.0000	0.0000
3670007	0.0916584248	0.0000	0.0000	0.0000
3932151	0.0982054712	0.0000	0.0000	-0.0000

Data process

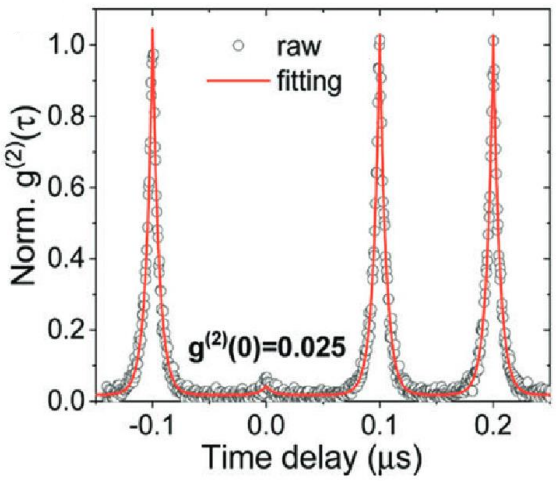
Snapi & Uniharp software g2 calculation

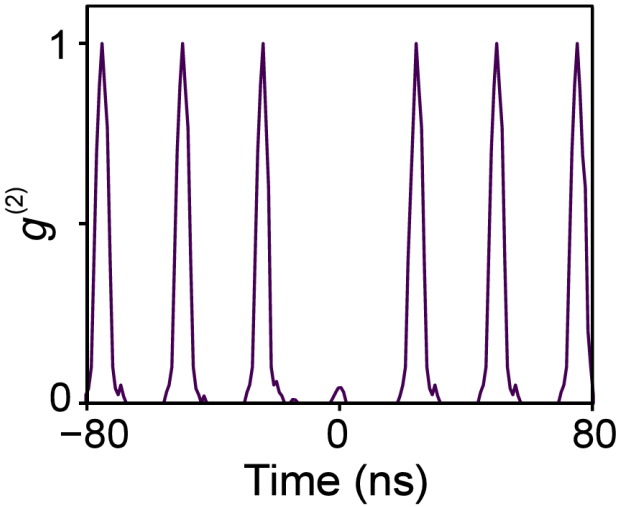


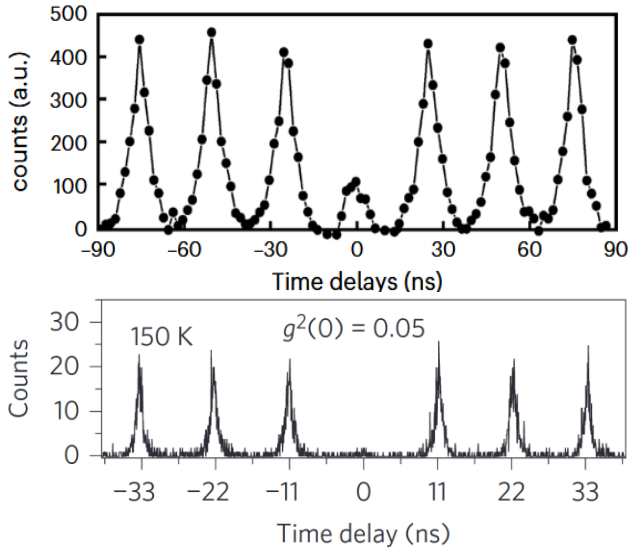


System setup

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Higher repetition rate:

- Higher time resolution and less pile-up effect
- Shorter integration time to get the same magnitude of measurement uncertainty

$$\mathrm{cps}_A \approx f_{\mathrm{rep}} p_A \qquad \mathrm{cps}_B \approx f_{\mathrm{rep}} p_B$$

$$C_0 \approx N_{\mathrm{pulse}}\, q_0 = (f_{\mathrm{rep}} T)\underbrace{(p_A p_B g^{(2)}(0))}_{q_0}$$

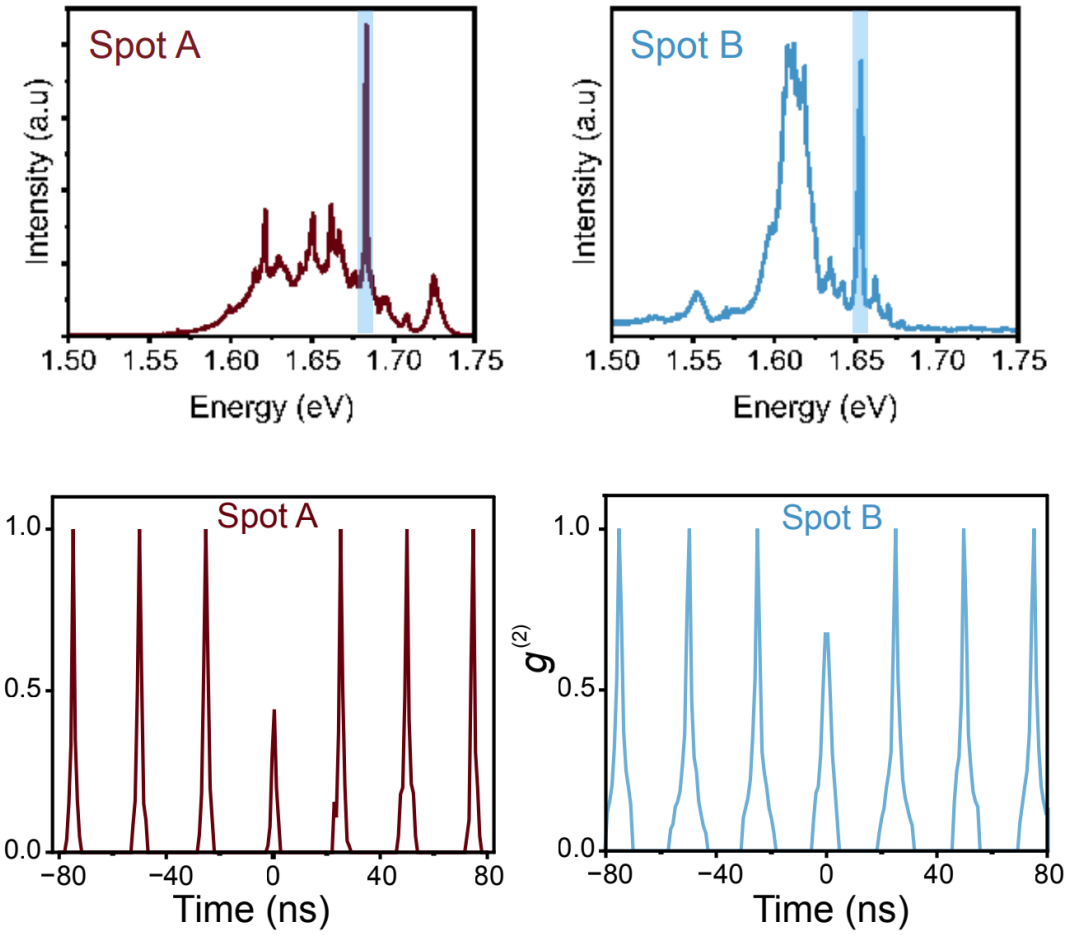
$$\mathrm{Var}(C_0) \approx C_0 \quad \frac{\sigma\left[g^{(2)}(0)\right]}{g^{(2)}(0)} \sim \frac{\sigma(C_0/C_\infty)}{C_0/C_\infty} \approx \sqrt{\frac{1}{C_0}}$$

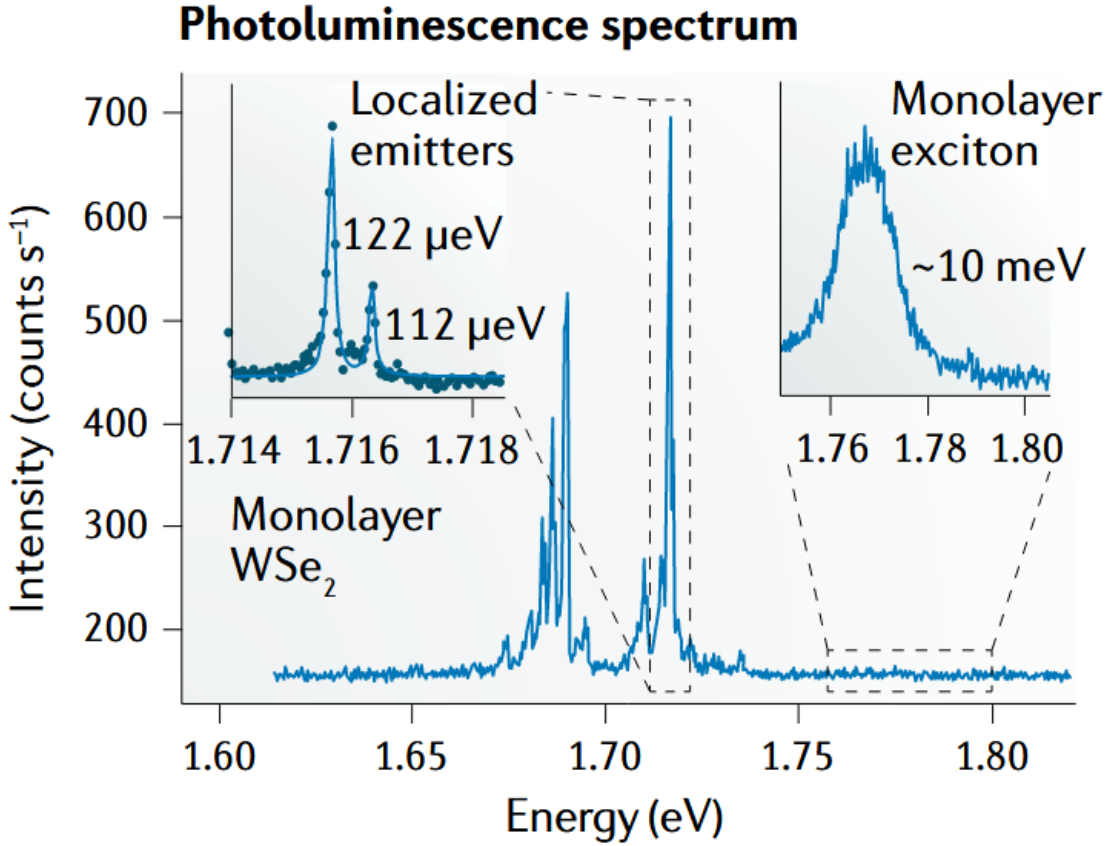
$$C_0 \gtrsim \frac{1}{\varepsilon^2} \Rightarrow T \gtrsim \frac{1}{\varepsilon^2 f_{\mathrm{rep}} p_A p_B g^{(2)}(0)}$$

Phys. Rev. Lett. 86, 1502

SPEs from localized defects

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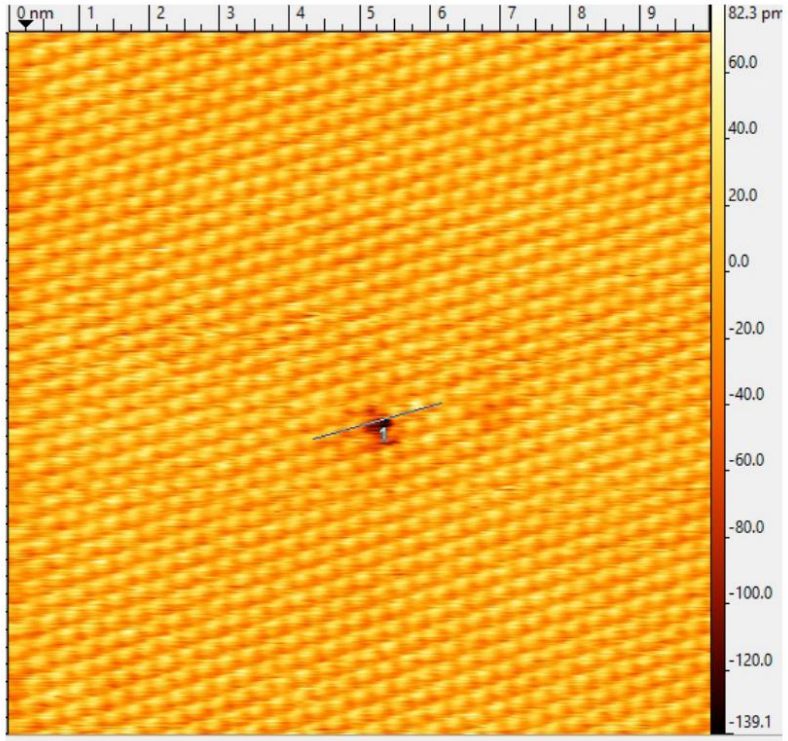


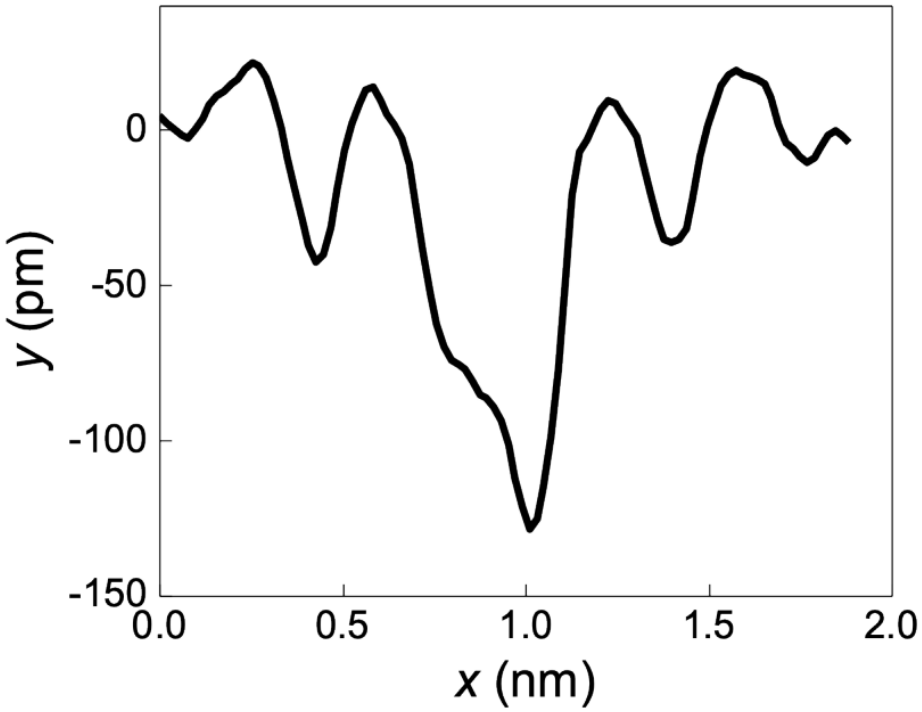


Nano Lett. 2025, 25, 11226−11233

Localized defect

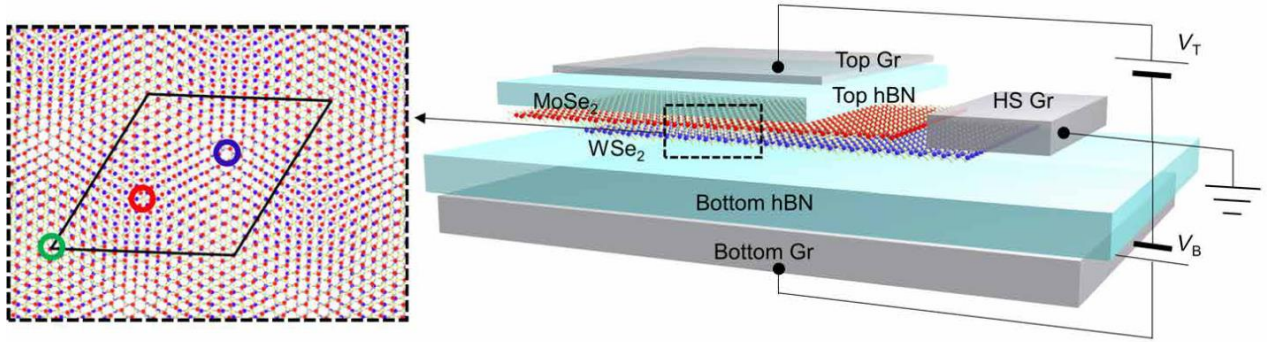
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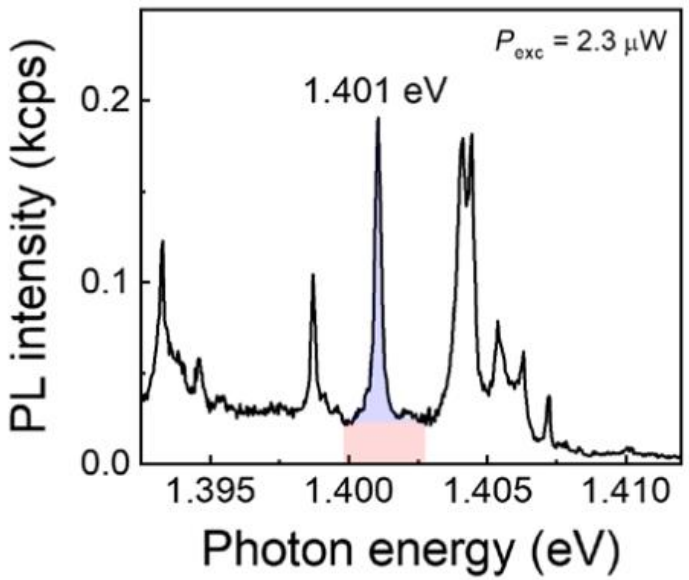


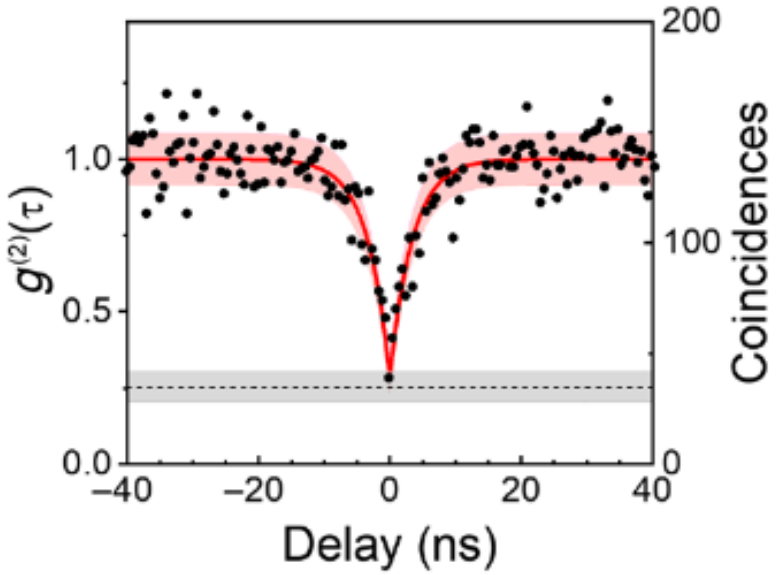


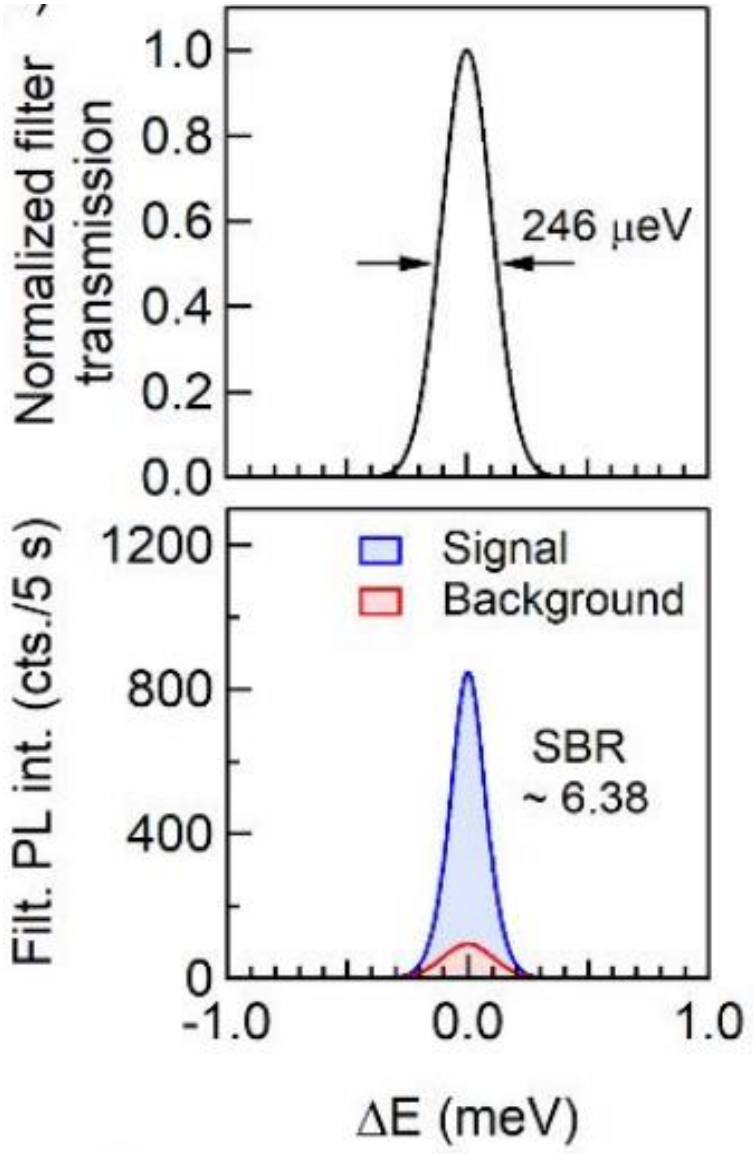
SPEs from moiré-trapped interlayer exciton

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Sci. Adv. 6, eaba8526 (2020)

TMD materials emission characteristics

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Exciton Class	Typical g²(0) Behavior
Neutral / Biexciton / Trion	≈ 1 — multimode emission, detector jitter
Localized (defect / strain)	< 0.5 — single-photon emission
Moiré-trapped interlayer exciton	< 0.5 (individual), ensemble ≈ 1
Exciton condensation emission	> 1 — photon bunching

$$g^{(2)}(\tau) = 1 + \frac{\langle \delta I(t)\ \delta I(t+\tau) \rangle}{\langle I \rangle^2} \qquad g^{(2)}_{QM}(0) = 1 + \frac{(\Delta n)^2 - \langle n \rangle}{\langle n \rangle^2}$$



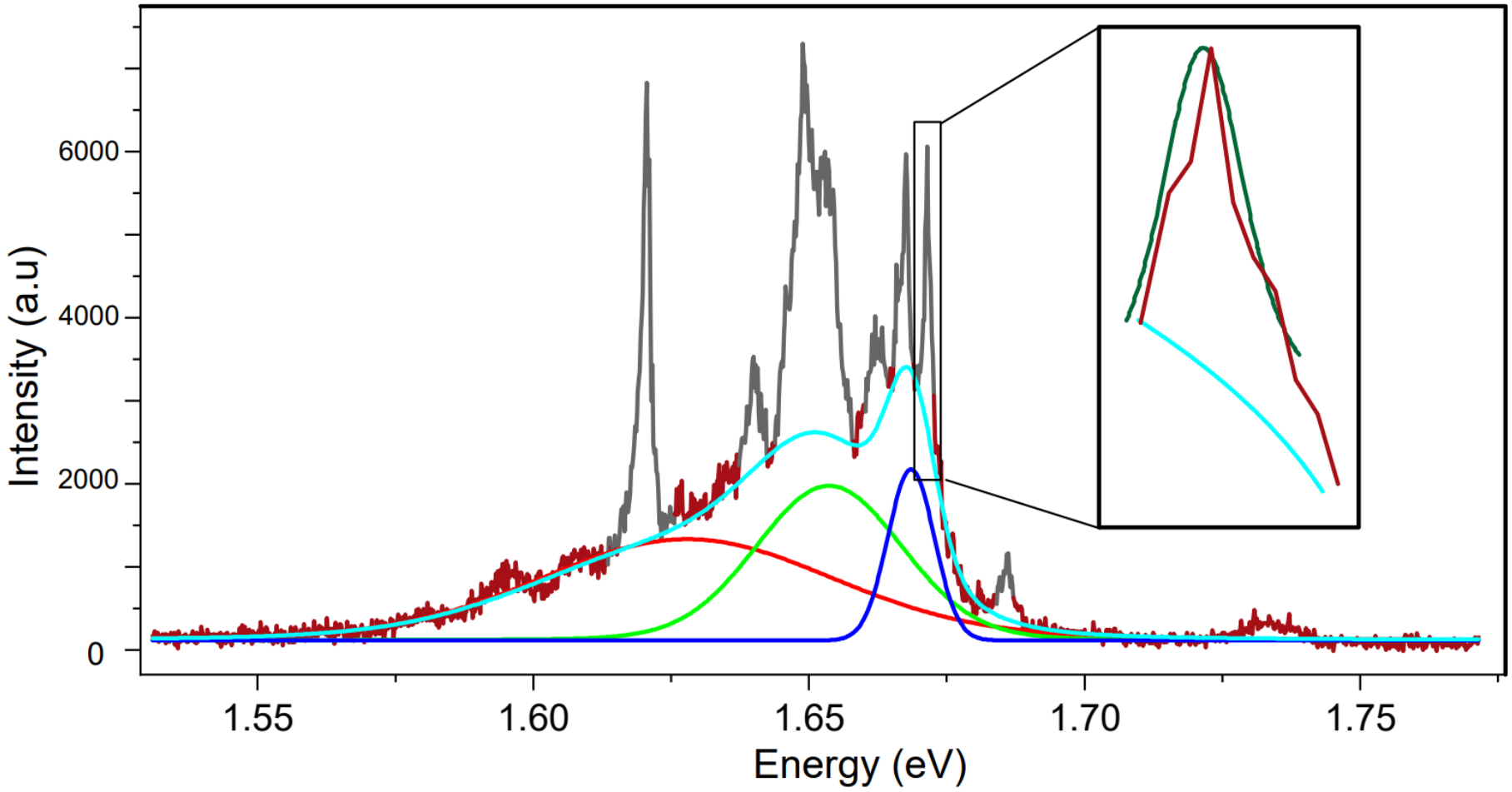
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THANKS

Speaker: Zhongqi Xiu

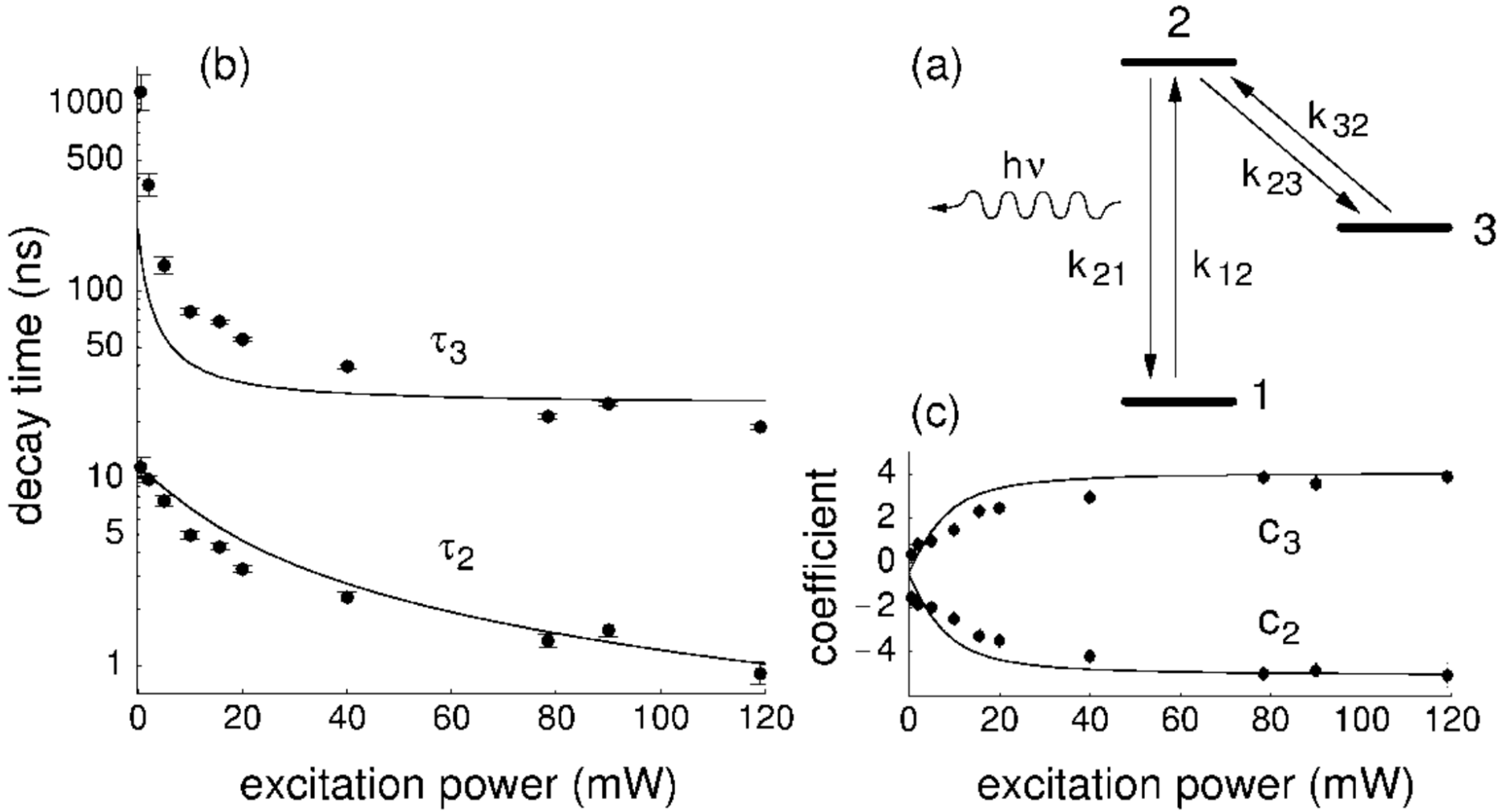
PL fit

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Three level system

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Setup and measurement

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